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Palaeomagnetism and the age of the Cracow volcanic rocks (S Poland)

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SUMMARY

Permian rhyodacites, melaphyres and tuffs from the Cracow area (South Poland) were sampled for the palaeomagnetic and isotope studies. Single-grain U-Pb dating of most zircon grains separated from the rhyodacites gave mean age of magma emplacement of 294.2 ± 2.1 Ma. Some zircons, however, displayed younger ages (268.7 ± 3.4 Ma), probably related to the metasomatic alterations of these rocks. Two Permian components of magnetizations related to these processes were isolated and together with previously defined Late Carboniferous–Permian palaeomagnetic poles from South Poland were used for construction of the regional apparent polar wander path (APWP). The Early Permian segment of this APWP shows a certain departure from the coeval part of the Fennoscandian APWP due to anticlockwise rotations of studied rocks most probably caused by mid-Permian sinistral tectonic movements along reactivated prominent Variscan faults of Central Europe. This sense of tectonic mobility does not support the hypothesis about transformation from Pangea 'B' to Pangea 'A' along an intra-Pangea dextral megashear during the Permian. Older than previously assumed ages of the post-Variscan igneous rocks of Central Europe reduce overlap of Gondwana's and Laurussia's parts of the Early Permian Pangea 'A'.

Key words: Palaeomagnetism applied to tectonics; Continental tectonics: strike-slip and transform; Europe.

1 INTRODUCTION

The Late Palaeozoic configuration and history of Pangea is still debated. Most of palaeomagnetic data indictate a substantial amount of overlap of continental crust in the Early Permian between Gondwana and Laurussia if reconstructed in Pangea 'A' configuration (Wegener 1924; Irving 1977, 2004; Muttoni *et al.* 1996). The introduction of the Pangea 'B' model (Irving 1977) seemed to eliminate this problem. Since Pangea 'A' is well defined in the Early Jurassic just prior to opening of the North Atlantic the scenario of transformation from the model 'B' to 'A' had to be presented. According to Muttoni *et al.* (2003) the transformation from Pangea 'B' to Pangea 'A' took place during the Permian because Late Permian palaeomagnetic data allow a Pangea 'A' configuration. They cited also geological factors supportive of an intra-Permian age for the transition. This transformation along an intra-Pangea dextral megashear system of about 3500 km occurred after the cooling of the Variscan mega-suture (see Arthaud & Matte 1977, Matte 1991) and lasted *ca*. 20 Myr. It should be stressed that in the model of Muttoni *et al.* (2003) the main SE–NW trending faults of Central Europe being a part of the Late Palaeozoic strike-slip system of Arthaud and Matte (1977) acted also as dextral ones.

Torcq *et al.* (1997) postulate Triassic age of the intra-Pangean translation. In their model the Early Triassic global transgression was a consequence of the development of new ocean ridges resulting from the rupture of Pangea B and the opening westward of NeoTethys.

New Permian palaeomagnetic data from Gondwana seem to support a Pangea 'B' palaeogeography in Early Permian times (Rakotosolofo *et al.* 2006). However, detailed selection of Permian palaeomagnetic poles from Laurussia performed by Van der Voo and Torsvik (2004) significantly decreases the overlap between both parts of Pangea 'A' model. Moreover, new isotope study of Permian igneous rock from Scandinavia (Van der Voo & Torsvik 2004) may indicate that the mean age of the Autunian volcanites from post-Variscan Europe can be older than previously assumed. If so, then overlap of Gondwana's and Laurussia's parts of the Early Permian Pangea 'A' could be additionally reduced.

The palaeomagnetic data from the Sudetes (Nawrocki 1998) and some geological observations performed in the southern part of Germany (Mattern 2001) indicate that during the Permian the NW–SE trending faults recorded rather a sinistral sense of the tectonic transport. Anticlockwise rotations of the Late Carboniferous to Early Permian palaeomagnetic poles with respect to the coeval poles from the Fennoscandia were reported from the Sudetes and because of this, a local Sudetian apparent polar wander path has been proposed (Nawrocki 1998; Jeleńska *et al.* 2003).

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The preliminary aim of this paper is to define an age of the Cracow volcanic rocks and to yield palaeomagnetic poles from these rocks. New palaeomagnetic studies should answer the questions if older palaeomagnetic data from Europe used for the construction of the Permian Pangea are properly time constrained. We are also going to check if anticlockwise rotations deducted from palaeomagnetic data from the Sudetes were a local phenomenon or took place in the zone of another main fault of central Europe—Cracow-Hamburg as well.

2 STUDY AREA AND PREVIOUS PALAEOMAGNETIC RESULTS

The border zone of the Brunosilesia and Małopolska terranes is defined by a narrow fault zone 'Cracow-Lubliniec' that probably represents a segment of much larger, transcontinental tectonic line 'Cracow-Hamburg' (Fig. 1a; Żaba 1999). This zone has Proterezoic foundations, and was active from earliest Palaeozoic through to Cenozoic times. The effects of polyphase activity of this fault control the structural features of the whole zone including areas with igneous rocks of the Krzeszowice region, where vertical faulting, related to the tectonic activity of the deep substratum, led to the formation of numerous fault macroblocks (op. cit.).

The Late Palaeozoic volcanic rocks of the Krzeszowice region are exposed over an area of approximately 200 km^2 about 20 km west of Cracow in Southern Poland (Fig. 1b). Composition of the volcanic rocks ranges from basaltic trachyandesites to rhyodacites with minor andesites and lamprophyres (Czerny & Muszyński 1997, 2000). Basaltic trachyandesites form several lava flows intercalated with pyroclastics and erupted along SE–NW trending faults bordering Nieporaz-Brodła graben (Fig. 1b). Sporadically basaltic rocks form small intrusions, for example, a sill in Niedźwiedzia Góra. Rhyodacites intruded at shallow levels forming predominantly laccoliths and have been encountered in Zalas-Orlej, Mikinia and Dubie vicinity. Felsic tuffs outcrop in the area of Filipowice. All volcanic rocks have high (or very high—e.g. Filipowice tuff) potassium contents. The potassium enrichment is attributed to postmagmatic processes, for example, adularyzation and seladonization (Słaby 1987, 1990; Parchoniak & Weiser 1992; Czerny & Muszynski 1997). Apart from the beds deformed in the contact zones developed near the igneous bodies the Permian and younger sedimentary rocks of the Krzeszowice region dips very gently (*ca*. 3°) to the east.

The first age determinations of volcanic rocks in the Krzeszowice region were based on the Filipowice rhyodacitic tuffs superposition relatively to travertines. The tuffs are contemporaneous with the travertines palaeontologically dated as being Early Permian in age (Lipiarski 1971). In turn, isotopic dates obtained for the Filipowice tuffs have been interpreted as timing of the late potassium remobilization. The K-Ar whole rock method yields 193 ± 0.7 Ma somewhat different from that of the biotite 230 ± 11 Ma (Borucki & Oberc 1964). Autigenic illite separated from the same rock showed 199.3 ± 0.7 Ma (Michalik *et al.* 2004). Dating of the Zalas rhyodacite with track's method indicate 295 Ma (Skowroński 1974). Basaltic trachyandesites have not been dated so far.

The exposures of the volcanic rocks of the Zalas-Orlej area are concordant with the gravimetric anomaly of a diameter of 4–6 km^2 (Kurbiel 1978). The study on the tectonics of the porphyritic intrusion proved that all the outcrops belong to a single magmatic stock (a laccolith), resulting from magma injection into the Lower Carboniferous mudstones (Dżułyński 1955). These are porphyritic rocks with plagioclase, biotite and altered hornblende phenocrysts and red or grey matrix composed predominantly of quartz, K-feldspar and apatite. The rocks were classified as rhyodacites (Czerny & Muszyński 1997; Lewandowska & Rospondek 2003; Musiał 1996; Muszyński 1995). The variation of the rock colour is ascribed to late or postmagmatic rock alterations (Musiał 1996; Słaby 1987). Similar in size and chemistry Mikinia rhyodacite do not show macroscopic diversity but the top of the volcanic body displays high potassium enrichment (e.g. Bolewski 1939; Słaby 1990). The phenocrysts assemblage additionally to that described from Zalas contains quartz. On the basis of geological (Parachoniak & Weiser 1992) and geochemical (Czerny & Muszyński 2000) observations, genetically associated with the Mikinia rhyodacite volcano are tuffs from Filipowice. These felsic tuffs are classified as trachytes (op. cit.) probably due to potassium enrichment (Michalik *et al.* 2004). Mafic rocks cropping out in the vicinity of Regulice have in part fluidal textures and miarolitic cavities. The pyroclastic material terminates each lava flow. Fine grained plagioclase (form labrador-andesine-oligoclase group) and pyroxene (from diopside-augite group) groundmass with few olivine (hypershen often altered to iddingsite) phenocrysts is a significant feature of the massive variety of these rocks (Czerny & Muszyński 1997). They are classified as basaltic trachyandesites (op. cit.).

The volcanics from the Krzeszowice region were palaeomagneticaly studied by Birkenmajer and Nairn (1964). Result of this study is not helpful for stratigraphic correlation of these rocks because it was based on limited number of samples only and the statisctics parameters are very poor. Moreover, the authors did not perform the palaeomagnetic contact tests. An attempt of the age estimation for the Zalas laccolith using the palaeomagnetic method was done by Nawrocki *et al.* (2005). For this purpose samples were taken from the rhyodacites and neighbouring Visean sediments cropping out in three localities. Two components of magnetization were isolated in the volcanic rocks and Visean sediments. The 'A' component, common for the greenish rhyodacites and Visean sediments (positive contact test), was carried by magnetite and was regarded to be primary (Nawrocki *et al.* 2005). The comparison of the mean inclination of this component with the expected (reference) stable European inclinations lead to conclusions that the Zalas laccolith was emplaced about 280 Ma ago. The second component 'B', characteristic for reddish rhyodacites, was carried by hematite and according to the authors (op. cit.) was recorded during the Late Permian (*ca*. 260 Ma) metasomatic alterations of these rocks. The palaeomagnetic poles calculated for the 'A' component is not in agreement with the corresponding Permian segment of the Baltic APWP. Nawrocki *et al.* (2005) draw a preliminary interpretation, suggesting that the disagreement was due to anticlockwise block rotations of the studied rocks, most probably connected with the Early Permian sinistral transtensional tectonic regime in the area of Central Europe. Up to this time poor age constrains for the Permian segment of the Baltic APWP and limited number of studied outcrops and samples pointed out the necessity of further palaeomagnetic and isotope studies of the volcanics from the Krzeszowice region.

3 MATERIAL AND METHODS

Fresh specimens of the Zalas rhyodacite for the zircon separations and single-grain U-Pb dating were collected in the southwestern part of the Zalas quarry *ca.* 15 m below the contemporary top of the laccolite and in the Orlej quarry, 20 meters northeast from the contact with the Lower Carboniferous clastics (Fig. 1b). Samples

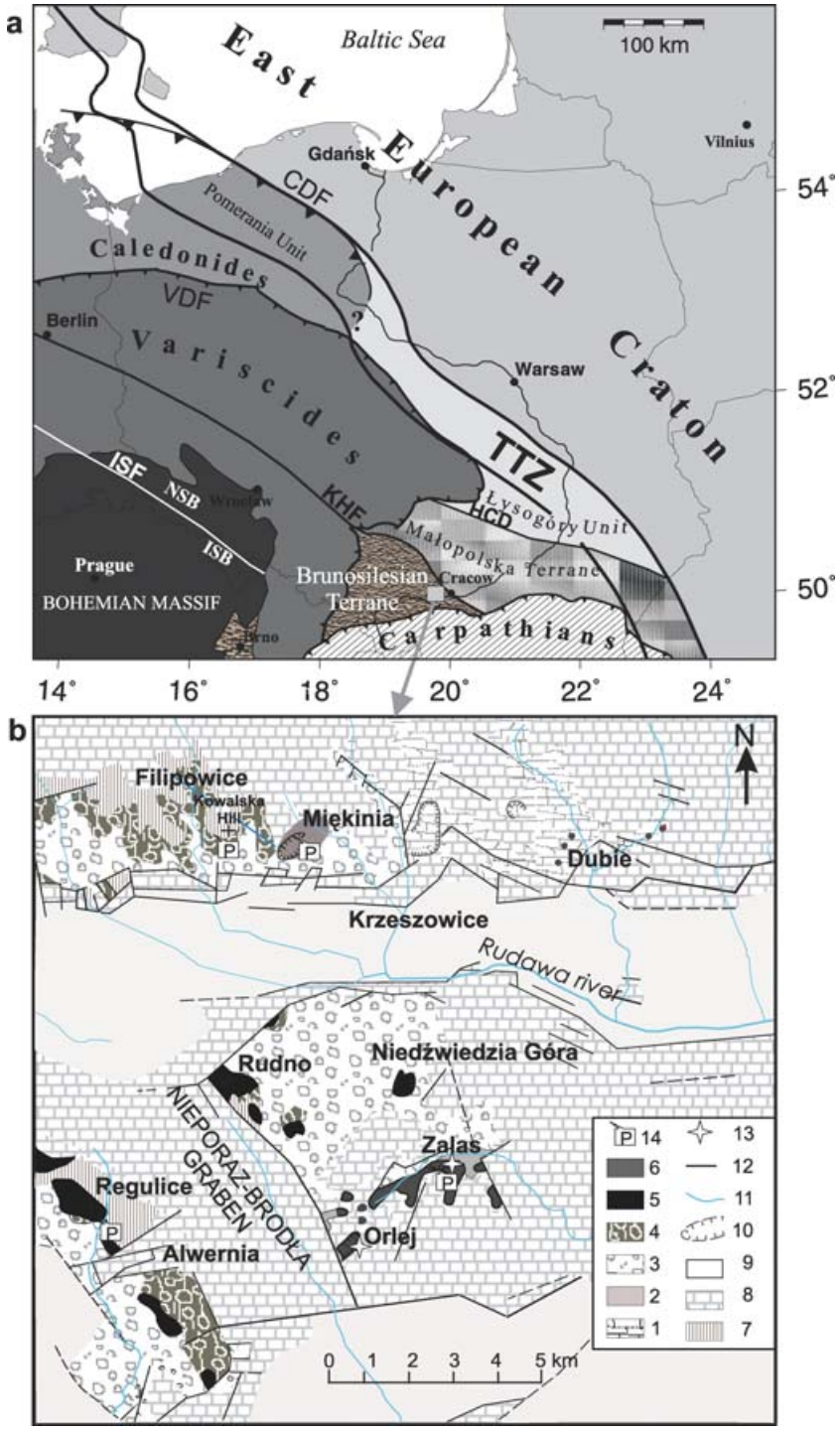


Figure 1. (a) Tectonic sketch map of Central Europe (after: Winchester and the PACE TMR NETWORK TEAM 2002, modified). TTZ – Teisseyre-Tornquist Zone, CDF – Caledonian Deformation Front; VDF – Variscan Deformation Front (after Pożaryski *et al.* 1992), HCD – Holy Cross Dislocation, KHF – Cracow-Hamburg Fault Zone, ISF – Inner Sudetic Fault, NSB – North Sudetic Basin, ISB – Intra Sudetic Basin. (b) Simplified geological map of Krzeszowice area (after Gradziński 1993): 1 – Devonian and Lower Carboniferous carbonates; 2 – Lower Carboniferous mudstones; 3 – Upper Carboniferous conglomerates; 4 – Permian conglomerates and travertines; 5 – Permo-Carboniferous basaltic trachyandesite lavas and sills; 6 – Permian rhyodacite domes and rhyolic, trachytic and andesitic dykes; 7 – Permo-Carboniferous volcanoclastics; 8 – Mesozoic sediments; 9 – Tertiary mudstones; 10 – quarries; 11 – rivers; 12 – faults; 13 – sites of U-Pb dating; 14 – sites of palaeomagnetic sampling.

were crushed, carbonates were removed with acetate buffer at pH 5, and heavy mineral fractions were separated using heavy liquid and magnetic separation techniques. From the heavy mineral fraction zircon grains were handpicked under a petrographic microscope. The separated zircons morphology was characterized using SEM HITACH S-4700. Twenty-two crystals were selected for single-grain U-Pb dating. The zircon population is very homogenous. All zircon crystals are euhedral, clear and colourless with smooth

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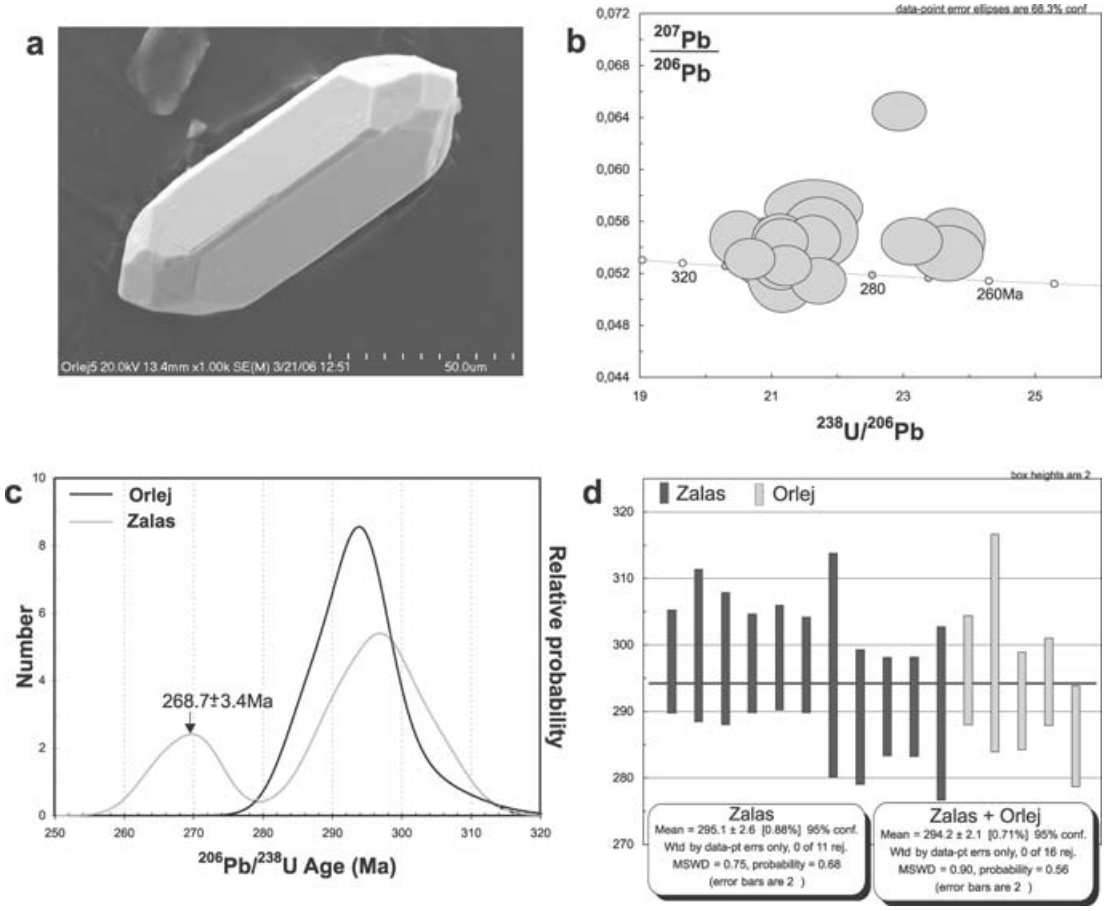


Figure 2. (a) The euhedral zircon crystal from Orlej separated for single crystal U-Pb dating. (b) Concordia diagram for 23 zircon grains from Orlej and Zalas. (c) Age relative probability diagrams prepared separately for the SHRIMP data from Orlej and Zalas localities. (d) Diagram of older population of U-Pb ages from Orlej and Zalas containing also error bars. The mean ages are also listed.

(Fig. 2a). They size reaches from 100 to 170 μm of length and 30 to 45 μm of width and the elongation ranges from 5:1 to 3:1. *The zircon grains were processed on the SHRIMP device* in the laboratory of the Research School of Earth Sciences, Australian National University.

The exposures of the rhyodacites from eastern part of the Zalas operating quarry and northern part of the Mikinia abandoned quarry (Fig. 1b) were sampled for the palaeomagnetic study. Samples for this purpose were taken also from felsic tuffs cropping out in the area of Filipowice (Kowalska Hill) and basaltic trachyandesites from the Regulice abandoned quarry. A total of 42 hand samples were collected from 13 sites of studied outcrops. Several core specimens 2.5 cm in diameter and 2.2 cm in length were drilled from each hand sample. Finally, 116 specimens were subjected to both alternating field (AF) and thermal demagnetization experiments. The remanences were measured with a Geofyzika JR5 spinner magnetometer. Demagnetization results were analysed using orthogonal vector plots (Zijderveld 1967), and the directions of the linear segments were calculated using principal component analysis (Kirschvink 1980). The studies of rock magnetic properties were performed in the Institute of Geophysics Polish Academy of Sciences in Warsaw, Poland and included the temperature dependence of bulk susceptibility at range (−192 to 700 °C)—with the KLY3-CS3 unit (Agico, Brno, Czech Republic), the temperature dependence of saturation isothermal remanence at range (20–700 °C)—with the laboratory made unit, the hysteresis parameters (hysteresis loops, Mr/Ms, Hc, Hcr)—with Micromag 2900 AGFM (Princeton Measurements, USA).

4 ISOTOPE AGES

A total of 23 analyses were done on single zircon grains (Table 1). Most of zircons from the Zalas quarry and all zircons from the Orlej quarry are concordant forming one cluster with an age of 294.2 ± 2.1 Ma (Figs 2b–d). The mean age calculated for the zircon grains from the Zalas quarry only is a bit older (295.1 ± 2.6 Ma). Surprisingly four zircon grains from the Zalas quarry assemble in the second cluster revealing the presence of another mean isotope age of 268.7 ± 3.4 Ma (Figs 2b and c). Analyses 16.1 (Table 1) is strongly discordant (Fig. 2b). In summary, the data from the Zalas rhyodacite provides strong evidences of the Early Permian (Sakmarian/Asselian) zircons, and clearly suggest also the presence of zircon grains of middle Permian age (Wordian-Kungurian). This age corresponds to the age of metasomatic alterations of these rocks inferred from preliminary palaeomagnetic data (Nawrocki *et al.* 2005).

5 ROCK MAGNETIC PROPERTIES

The thermomagnetic SIRM (T) curve for Zalas red rhyodacite (ZL 1) indicates that three ferromagnetic phases were present in fresh sample—with blocking temperatures of about 180 °C (low temperature 'LT' phase), 670 °C (hematite) and probably around

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Table 1. Summary of SHRIMP U-Pb results for zircons from Zalas and Orlej.

							Total			Radiogenic age			(Ma)	
Grain. spot	U (ppm)	Th (ppm)	Th/U	$^{206}Pb^{*}$ (ppm)	$^{204}Pb/^{206}Pb$	f_{206} %	$^{238}U/^{206}Pb$	±	$^{207}Pb/^{206}Pb$	±	$^{206}Pb/^{238}U$	±	$^{206}Pb/^{238}U$	±
ZALAS														
1,1	227	73	0.32	9.3	–	0.28	21.1	0.277	0.0546	0.0013	0.0472	0.0006	297.6	3.9
2,1	191	105	0.55	8.0	–	0.26	20.47	0.282	0.0546	0.0014	0.0487	0.0007	306.6	4.2
3,1	456	158	0.35	17.1	0.0012	1.59	22.92	0.268	0.0644	0.0010	0.0429	0.0005	271.0	3.2
4,1	198	96	0.48	7.2	0.0002	0.24	23.65	0.356	0.0535	0.0014	0.0422	0.0006	266.3	4.0
5,1	122	73	0.60	5.0	–	0.14	20.97	0.404	0.0535	0.0018	0.0476	0.0009	299.9	5.7
6,1	90	91	1,01	3.7	0.0012	<0.01	21.15	0.355	0.0519	0.0020	0.0473	0.0008	298.0	5.0
7,1	288	135	0.47	11.7	–	0.26	21.13	0.266	0.0544	0.0012	0.0472	0.0006	297.3	3.7
8,1	223	112	0.50	9.1	–	0.03	21.12	0.280	0.0526	0.0013	0.0473	0.0006	298.2	3.9
9,1	186	101	0.54	6.8	0.0008	0.36	23.71	0.337	0.0545	0.0016	0.0420	0.0006	265.4	3.8
10,1	381	351	0.92	15.4	–	0.03	21.2	0.259	0.0525	0.0010	0.0472	0.0006	297.0	3.6
11,1	256	85	0.33	10.4	0.0007	0.18	21.17	0.607	0.0537	0.0012	0.0472	0.0014	297.0	8.4
12,1	117	77	0.65	4.6	0.0003	0.37	21.71	0.382	0.0550	0.0018	0.0459	0.0008	289.2	5.1
13,1	346	275	0.80	14.4	0.0001	0.07	20.65	0.254	0.0530	0.0010	0.0484	0.0006	304.6	3.7
14,1	271	175	0.65	10.7	–	<0.01	21.69	0.278	0.0513	0.0012	0.0461	0.0006	290.8	3.7
15,1	266	71	0.27	10.6	–	0.30	21.61	0.280	0.0545	0.0012	0.0461	0.0006	290.8	3.7
16,1	1526	2092	1,37	44.9	0.0071	0.14	29.24	0.483	0.1680	0.0084	0.0291	0.0009	185.1	5.5
17,1	285	174	0.61	10.6	0.0002	0.33	23.11	0.306	0.0544	0.0012	0.0431	0.0006	272.2	3.6
18,1	195	90	0.46	7.8	0.0003	0.60	21.62	0.491	0.0569	0.0015	0.0460	0.0011	289.7	6.5
ORLEJ														
1,1	224	104	0.47	9.1	0.0008	0.46	21.16	0.294	0.0559	0.0014	0.0470	0.0007	296.3	4.1
2,1	266	131	0.49	10.9	0.0004	0.27	20.91	0.576	0.0545	0.0012	0.0477	0.0013	300.3	8.2
3,1	347	340	0.98	13.8	0.0003	0.07	21.59	0.274	0.0527	0.0011	0.0463	0.0006	291.6	3.7
5,1	807	252	0.31	32.4	0.0001	0.01	21.39	0.241	0.0523	0.0007	0.0467	0.0005	294.5	3.3
6,1	225	149	0.66	88	–	0.22	21.97	0.292	0.0538	0.0013	0.0454	0.0006	286.3	3.8

Notes: (1) Uncertainties given at the one σ level, (2) error in Temora reference zircon calibration was 0.79 per cent for the analytical session (not included in above errors but required when comparing data from different mounts), (3) f_{206} % denotes the percentage of ^{206}Pb that is common Pb and (4) correction for common Pb made using the measured $^{238}U/^{206}Pb$ and $^{207}Pb/^{206}Pb$ ratios following Tera & Wasserburg (1972).

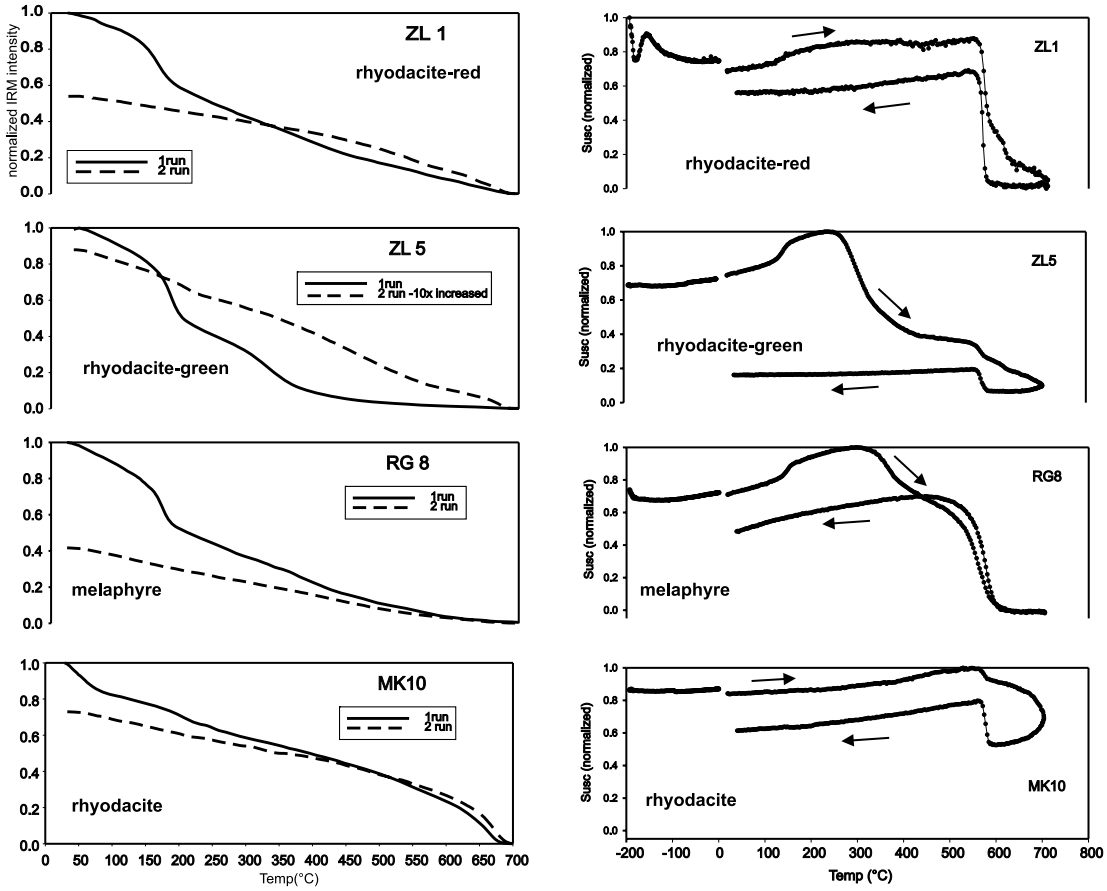


Figure 3. The thermomagnetic curves for selected samples: SIRM(T) – left-hand diagrams – solid curve – heating of fresh sample, dashed curve – second heating of the same sample. Intensity normalized for the maximum of solid curve; $K(T)$ – right-hand diagrams. – three curves are drawn – for heating from –184 °C to room temp and heating and cooling curve (marked with arrows) for 20–700 °C range. All experiments were performed in air atmosphere. Some redox conditions are probable for heating powdered sample at $K(T)$ experiment.

580 °C (magnetite) (Fig. 3, left-hand side). The sample previously heated to 700 °C does not show the presence of 180 °C phase. The susceptibility curve (Fig. 3, right-hand side) reveals the possible Verwey transition (typical for magnetite), magnetite at 580 °C and a tail for hematite. At the range of 100–250 °C slight increase in susceptibility may indicate the mineralogical change of the low temperature phase. Upon heating to 700 °C hematite is reduced and magnetite persists. The high value of coercivity of remanence (Hcr of 80 mT, Fig. 4a), moderately steep IRM acquisition curve (Fig. 4b) and coercivity (30 mT, Table 2) also support the magnetite—hematite grains mixture. The hysteresis loop is controlled by the ferromagnetic phase. For Zalas green rhyodacite (ZL5) phases with blocking temperatures of 180 and 350 °C dominate the IRM signal (Fig. 3). The heating to 700 °C results in the generation of hematite and magnetite and growth of SIRM by almost 10 times probably due to oxidation of some iron sulphides (pyrrhotite, pyrite?). The $K(T)$ susceptibility curve show substantial mineralogical changes of LT phase, slight presence of magnetite and the reduction of hematite at the temperature range of 580–700 °C to less susceptible phase—during heating and cooling modes. The cooling curve shows only the presence of magnetite. Hc and Hcr parameters are lower than for ZL1 supporting that conclusion that hematite grown upon heating of SIRM is secondary. In Mikinia red rhyodacite (MK10) both SIRM(T) and $K(T)$ experiments reveal the dominating hematite phase. Some LT phase (with Tb $<$ 100 °C) is observed on SIRM (T) curve (Fig. 3). Little magnetite is also observed on the $K(T)$ curve. Heating of powdered sample at 600–700 °C leads to the reduction of some hematite (with very high Curie temperature). The coercivity value Hc (28.46 mT) and high values of coercivity of remanence (Hcr $>$ 300 mT) indicate that mixture of ferrimagnetic grains is dominated by hematite with some minor magnetite content (seen at IRM acquisition curve, Fig. 4). For Regulice melaphyre SIRM(T) and $K(T)$ data indicate the LT phase as well as magnetite without any hematite presence. Hysteresis data also are typical for samples dominated by magnetite with steep slope of IRM acquisition curve, very low Hc (7–22 mT) and Hcr (24–29 mT). Day diagram indicates mostly PSD behavior for samples with the substantial contribution of magnetite (ZL1, ZL5, RG8). For hematite–rich MK10 rhyodacite the Hcr/Hc ratio is typical for hematite (with high Hcr values). The 'LT' phase with low blocking temperatures and undergoing mineralogical changes upon heating above 150–180 °C can be some Ti-poor titanomagnetite, maghemite or monoclinic pyrrhotite.

6 PALAEOMAGNETIC RESULTS

One distinct component of NRM labelled 'A', with blocking temperatures not exceeding 600 °C and inclination not higher than −20°, occurs in the greenish rhyodacites, part of melaphyre samples, and all samples of tuff (Fig. 5). Reddish rhyodacites and most of melaphyre samples contain another component labelled 'B' with high coercivity and blocking temperatures exceeding 600 °C, and inclination enclosed between −20 and −35°. Both components are

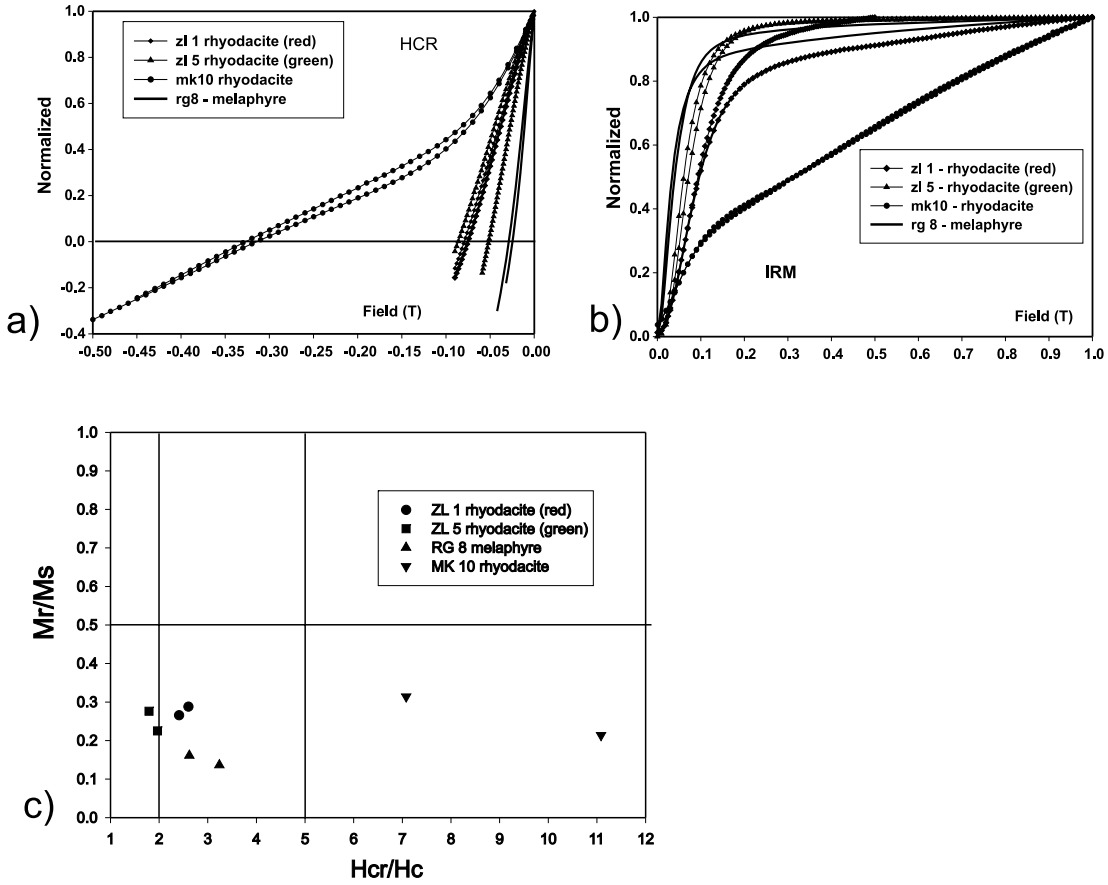


Figure 4. (a) The hysteresis properties of selected samples – recorded with Micromag AGFM 2900. Two subsamples were measured for each sample. (b) Coercivity of remanence with saturating field of 1 T, IRM acquisition curves. (c) The Day plot for Hcr/Hc and Mr/Ms ratios.

Table 2. The hysteresis parameters for four samples (with two subsamples measured).

Sample	Ms norm (mA m^2 kg^{-1})	Mr norm (mA m^2 kg^{-1})	Mr/Ms	Hc (mT)	Hcr (mT)	Hcr/Hc
Zalas rhyodacite (red)						
zl 1 a 0.5 T	61.03	17.58	0.29	30.48	79.28	2.60
zl 1 b 1 T	59.95	15.93	0.27	31.18	75.09	2.41
Zalas rhyodacite (green)						
zl 5 a 1 T	212.70	47.88	0.23	21.87	43.01	1.97
zl 5 b 1 T	360.00	99.40	0.28	28.75	51.55	1.79
Regulice melaphyre						
rg 8 a 1 T	120.50	16.44	0.14	7.60	24.61	3.24
rg 8 b 1 T	297.10	47.88	0.16	11.00	28.85	2.62
Miękinia rhyodacite						
mk 10 a 1 T	9.64	2.06	0.21	28.38	314.60	11.09
mk 10 b 1 T	6.79	2.13	0.31	46.24	327.30	7.08

Note: Determined with AGFM Micromag 2900—maximum field of 0.5 and 1 T, hysteresis curve corrected for linear part.

grouping well in the third quarter or between the second and third quarters of the hemisphere (Fig. 6). The mean directions calculated for the component 'A' in particular localities have very consistent shallow inclination and good Fisher's statistics parameters (Table 3). The mean direction calculated for the component 'B' is characterized by inclinations about 15° higher and also good Fisher's statistic parameters. After taking into consideration of previous data from the Zalas laccolith (Nawrocki *et al.* 2005) the mean directions were calculated for the whole Cracow area (Table 3). The average direction 'A' does not contain results from the Kowalska Hill, where the mean declination of component 'A' differs of about 20° from the mean declinations of this component calculated for the rest of localities.

7 INTERPRETATION AND DISCUSSION

The inclination of component 'A' isolated in greenish rhyodacites, melaphyres and tuffs of the Cracow region ($-13^\circ \pm 4.8^\circ$), carried mainly by magnetite, correspond strictly to the inclination characteristic of the Visean sediments from the contact zone of the Zalas laccolith ($-13^\circ \pm 6.2^\circ$), where synfolding remanence attributed to the magma emplacement was defined (Nawrocki *et al.* 2005). The

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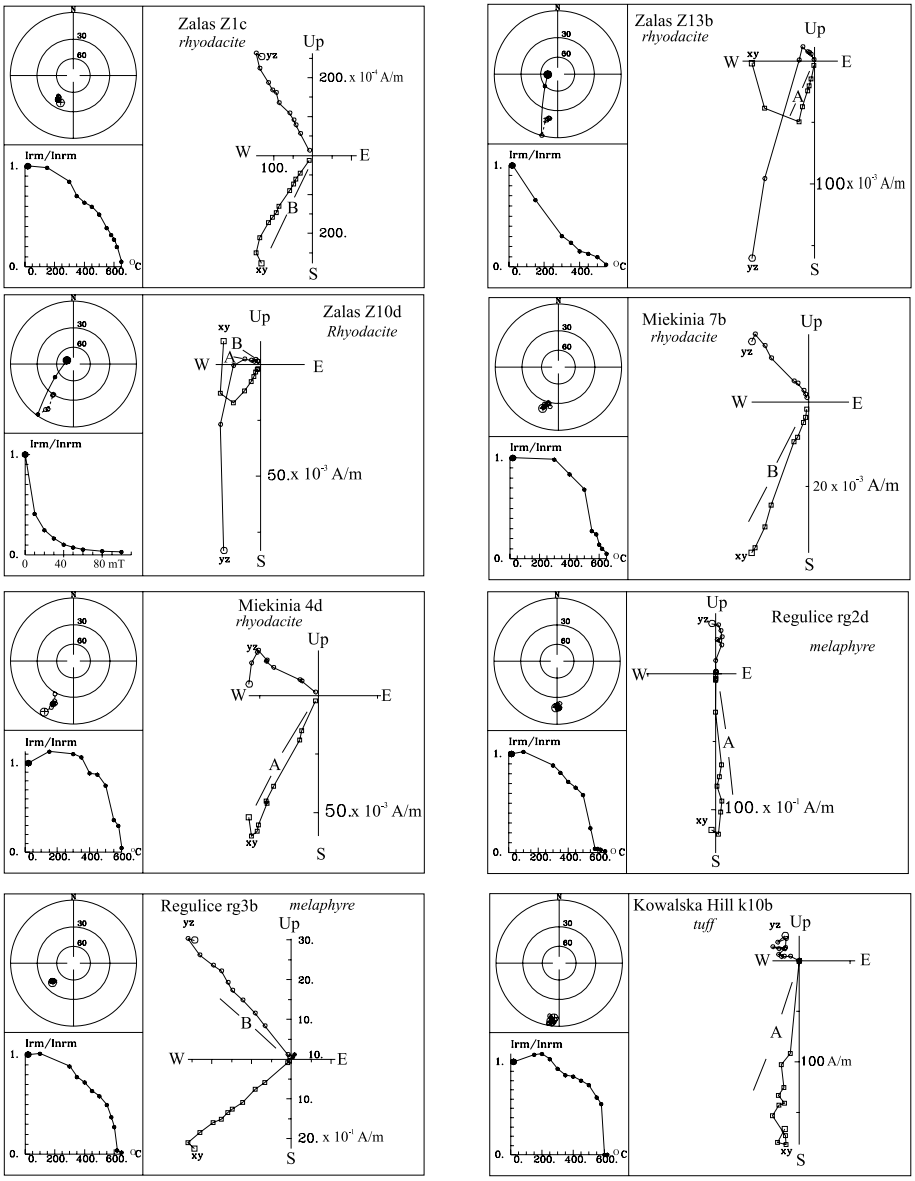


Figure 5. Typical demagnetization characteristics (demagnetization paths, intensity decay curves and orthogonal plots) of‘ Early Permian igneous rocks and tuffs from Cracow area. Circles in the orthogonal plots represent vertical projections, squares represent horizontal projections. Irm – intensity of remanent magnetization, Inrm – initial intensity of natural remanent magnetization. The diagrams were prepared by the means of computer package written by Lewandowski *et al.* (1997).

majority of zircon grains from the localities Zalas and Orlej give the mean age of about 294 Ma (Fig. 2d) defining the time of magma emplacement. Because of this the component ‘A’ was recorded *ca*. 294 Ma ago. The second component, ‘B’, characteristic for the reddish rhyodacites and melaphyres that underwent metasomatic alterations, is carried by hematite and was recorded most probably thanks to metasomatose. There is no direct evidence that the younger zircon age (Fig. 2b, Table 1) is coeval with metasomatic alterations. It should be stressed, however, that there is no other processes that could reset or annex zircon grains in the Cracow volcanites. Because of this the most likely age of the metasomatose and the component ‘B’ is of about 270 Ma (Fig. 2c).

The mean palaeomagnetic poles calculated for the Cracow volcanites have been compared with the apparent polar wander path (APWP) characteristic for the Fennoscandian part of the East European Craton. The APWP characteristic for the whole of pre-Alpine Europe should be treated rather as to big simplification since many Late Carboniferous to Early Permian poles from Variscan Europe depart distinctly from the coeval Baltic poles (Cogné *et al.* 1993; Nawrocki 1998; Henry *et al.* 1999; Jeleńska *et al.* 2003). It should be stressed that these poles display the same, anticlockwise course of rotations. The departure of the poles defined in the Permian St Affrique and Largentiere basins is explained as resulting from regional syn-extensional tectonics (Cogné *et al.* 1993; Henry *et al.* 1999). Anticlockwise rotations of palaeopoles isolated from the Late Carboniferous to Early Permian sedimentary and volcanic rocks of Sudetes is regarded as consequence of regional transtension and activity of sinistral faults (Nawrocki 1998) or as effect of block rotation of the whole Variscan Europe (Jeleńska *et al.* 2003).

The Late Carboniferous to Triassic segment of the Fennoscandian APWP (Fig. 7) has been constructed on the basis of the best quality

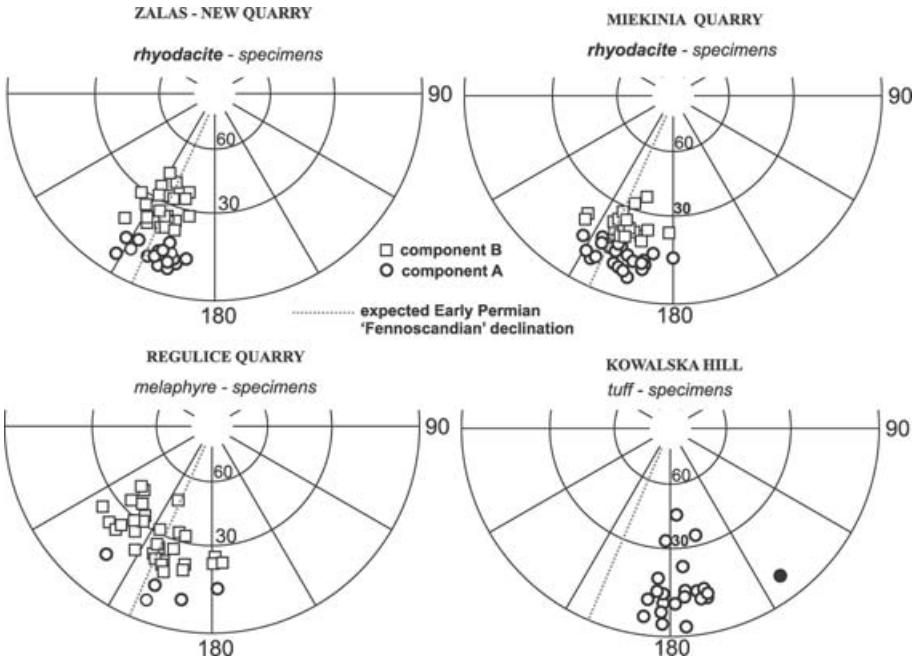


Figure 6. Stereographic projections of line-fit palaeomagnetic directions from the studied volcanic rocks. Open (closed) symbols denote upward (downward) pointing inclinations.

Table 3. Statistics of palaeomagnetic data obtained from the volcanogenic rocks of the Cracow region (50.1°N, 19.7°E).

Rock and locality	Com.	*n*/*N*	*D*	*I*	α_{95}	*K*	Plong.	Plat.	d*p*	d*m*	References
Rhyodacites, Zalas New Quarry (E part)	A	17/6	203	−13	5.9	127.9	347°E	43°S	3	6	This paper
	B	21/8	204	−27	3.9	202.1	342°E	50°S	2	4	This paper
Rhyodacites, Miekinia Quarry	A	27/8	198	−14	6.4	75.2	353°E	45°S	3	7	This paper
	B	26/10	201	−26	4.3	129.8	346°E	50°S	2	5	This paper
Melaphyres, Regulice Quarry	A	5/3	195	−12	25.1	24.8	358°E	44°S	13	25	This paper
	B	35/10	208	−28	8.6	32.7	337°E	48°S	5	9	This paper
Tuffs, Kowalska Hill	A	23/11	175	−15	5.6	30.6	27°E	47°S	3	6	This paper
Rhyodacites, Zalas Old Quarry	A	14/5	201	−14	4.3	319.2	350°E	43°S	2	4	Nawrocki *et al.* (2005)
Rhyodacites, Orlej Quarry	A	16/5	190	−14	6.8	128.1	5°E	46°S	4	7	Nawrocki *et al.* (2005)
	B	8/4	197	−26	7.2	162.3	353°E	51°S	4	8	Nawrocki *et al.* (2005)
Rhyodacites, Zalas New Quarry (N part)	B	7/3	206	−31	14.6	59.4	338°E	51°S	8	15	Nawrocki *et al.* (2005)
Cracow volcanites (rhyodacites and melaphyres) – summary statistics	A	5 loc.	197	−13	4.8	256.1	355°E	44°S	2	5	
	B	5 loc.	203	−28	4.1	345.1	344°E	50°S	2	5	

Note: n/*N* – number of specimens and number of samples used in final statistics, *D* – declination, *I* – inclination, α_{95}, *K* – Fisher's statistics parameters, Plat. – geographic palaeolatitude of south palaeopole, Plong. – geographic palaeolongitude of south palaeopole, d*p* – palaeodeclination error, d*m* – error of the distance between the site and palaeopole.

poles with Q factor (see Van der Voo 1993) not less than four (Table 4). They were taken from *the Global Palaeomagnetic Database* (http://www.ngu.no/dragon/) and time constrained according to Van der Voo and Torsvik (2004). The palaeopole 'B' is convergent with the Middle–Late Permian segment of the reference APWP while the palaeopole 'A' departs from the Early Permian segment of this APWP to the east (Fig. 7).

Similar eastern direction and magnitude of departure was observed for most of palaeomagnetic poles isolated from the Late Carboniferous–Early Permian rocks in the Sudetes (Nawrocki 1998; Jeleńska *et al.* 2003). Because of this a separate APWP characteristic for S Poland is proposed (Fig. 7). It is based on the palaeomagnetic poles from Sudetes, Cracow region and the Holy Cross Mountains (Table 4). This APWP overlaps the Fennoscandian APWP since *ca*. 270 Ma. The magnitude of rotation of Early Permian palaeomagnetic poles isolated from igneous rocks from the Cracow region and the Sudetes with respect to the Fennoscandian coeval poles is of about 15°. The amplitude of the rotation is higher when we compare the Early Permian poles isolated from the Early Carboniferous and Lower Permian sedimentary rocks covering the Permian intrusive bodies. The biggest departure displays the pole 'KH' isolated from the tuffs of Kowalska Hill and the pole 'OC2' characteristic for

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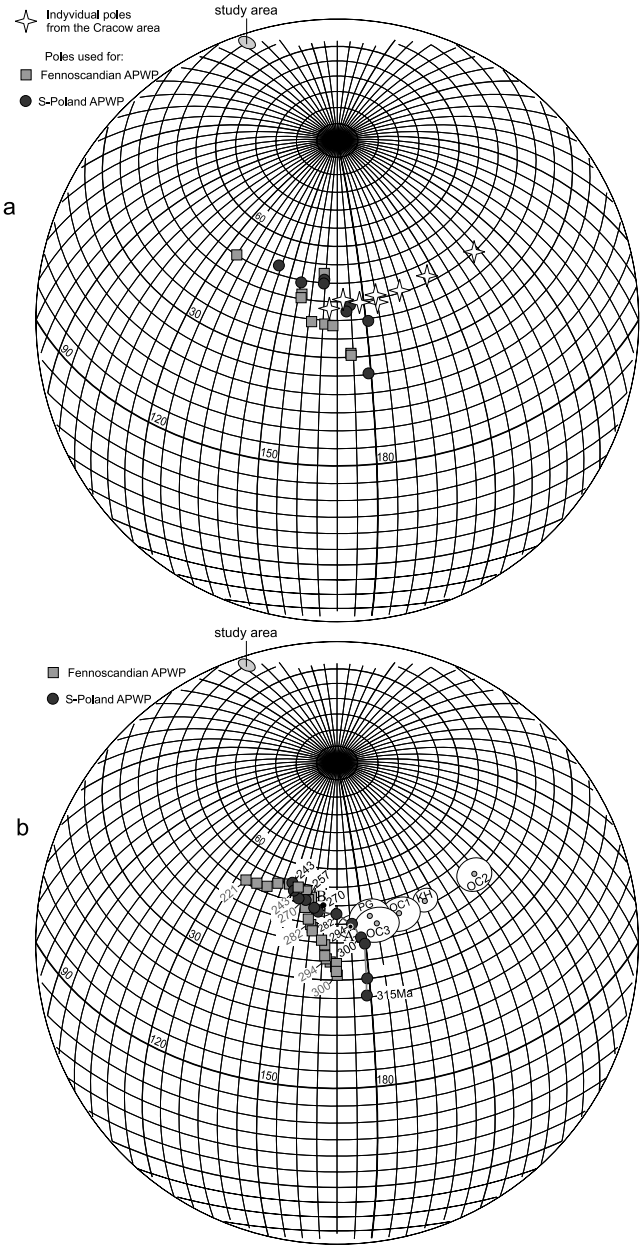


Figure 7. (a) Location of palaeomagnetic poles used for the construction of Fennoscandian and S Poland APWP (Table 4), and individual palaeomagnetic poles (calculated for the component A) from the Cracow area. (b) The Late Carboniferous to Middle Triassic apparent polar wander paths for Fennoscandia and southern Poland. They are based on the poles listed in Table 4. For the construction of paths the GMAP package with spline method (Torsvik & Smethurst 1992) was used. Selected Permian palaeopoles (with 95 per cent confidence ovals) from sedimentary rocks of Cracow area and Sudetes are also presented. OC1, OC2, OC3—palaeopoles from the Visean sediments covering southern part of Zalas intrusion at Orlej (Nawrocki *et al.* 2005); KH – palaeopole from tuffs of Kowalska Hill (this paper); PG – palaeopole from Płóczki Górne sediments (Upper Autunian, North Sudetic Basin) (Nawrocki 1998).

the Visean mudstones covering southern part of the Zalas intrusion (Fig. 7).

Up till now an age of the Autunian volcanites from the Sudetes have not been constrained by any modern isotope study. In order to define their palaeomagnetic age, the palaeoinclinations defined for some Sudetian volcanic rocks (Nawrocki 1997, 1998), after recalculation to geographic coordinates of Cracow, have been compared with the reference curve of palaeoinclination characteristic for Late Carboniferous–Late Permian of southern Poland. It is clearly visible that inclination of direction 'A' characteristic for the Cracow volcanites and inclinations characteristic for the Autunian rhyodacites and melaphyres studied in the Sudetes form one cluster (Fig. 8). Therefore, it is likely that Autunian volcanic activity in both places of southern Poland took place *ca.* 294 Ma ago and was constrained to a relatively short geological event.

The influence of the Early Permian magmatism on the palaeomagnetic record in remagnetized carbonates of western and central Europe still has not been proven. Permian remagnetizations of Palaeozoic carbonates were documented in several places of pre-Alpine Europe (e.g. Nawrocki 1993; Zegers *et al.* 2003; Zwing *et al.* 2005; Grabowski *et al.* 2006). A comparison of reference curve of palaeoinclination with the inclinations characteristic for remagnetized Devonian carbonates from Upper Silesia and Holy Cross Mountains (Nawrocki 1993; Grabowski *et al.* 2006) leads to the conclusion that two events of Permian remagnetization of these carbonates were coeval with the magma emplacement and with the metasomatose of volcanic rocks from the Cracow region (Fig. 8). It means that remagnetizing brines were most probably initiated by the same processes. The paths for regional migration of brines and thermochemical or chemical remagnetizations were opened thanks to two Permian tectonic events. A limited time of Autunian magmatism in both regions of southern Poland and observed rotations of Early Permian palaeomagnetic poles could indicate that the magma emplacement and its later metasomatose were accompanied by a short periods of transtensional tectonic activity.

The eastern departures of northern palaeomagnetic poles isolated from the Autunian rocks of southern Poland may indicate anticlockwise rotations of the studied rocks of magnitude up to 45° (pole OC1, Nawrocki *et al.* 2005). The youngest pole deviated in such a way from the Fennoscandian reference path was isolated from the upper Autunian sediments covering the Early Permian igneous rocks in the North Sudetic Basin (Nawrocki 1998). On the other hand the palaeomagnetic pole 'B' and pole isolated from the Zechstein sediments of Sudetes (Nawrocki 1997) overlap the Fennoscandian APWP. Because of this, the onset of anticlockwise tectonic rotations predates Zechstein and metasomatic alterations of Cracow volcanics.

The mid-Permian anticlockwise rotations of the studied rocks were inducted by wrenching tectonics, most probably associated with a regional sinistral strike slip activity of the Cracow-Hamburg prominent fault zone that separates two Cadomian terranes—Brunosilesia and Małopolska. The same process was proposed for tectonic rotations of the same age and direction detected in the Sudetes (Nawrocki 1998) where sinistral tectonic activity revealed at that time the Intra-Sudetic Fault. Some structural studies (Żaba 1999; Aleksandrowski *et al.* 1997) support our hypothesis about Permian sinistral tectonic activity of main fault zones in southern Poland. Another solution assuming regional tectonic anticlockwise rotation of the whole Variscan belt in Poland (Jeleńska *et al.* 2003) cannot be inferred from palaeomagnetic data because the magnitude of rotations is different being usually less in the igneous rocks than in their sedimentary cover. Moreover, this process involved also the Cracow area that was outside of the Variscan orogen. The Early Permian left-lateral motion between Baltica and Western Europe recorded by major NW-trending faults of the Bohemian Massif is postulated by Mattern (2001) who interpreted a strike-slip origin of some Early Permian basins.

The sinistral movements along major NW-trending faults cutting the Cracow area, Sudetes and the Bohemian Massif (Fig. 9a)

Table 4. Palaeopoles used for construction of apparent polar wander paths and presented separately in Fig. 7.

Rocks, locality	α_{95}	PLAT	PLONG	Mean age (Ma)	*Q*	References
Fennoscandian part of East European Craton						
Mt Billinger Sill, Sweden	4.0	31	174	299	5	Mulder (1971)
Ny-Helesund Diabases, Norway	2.9	39	161	296	4	Halvorsen (1970)
Scania Dolerites, Sweden	6.5	38	168	294	5	Mulder (1971)
Stabben Sill, Norway	2.4	32	174	291	4	Sturt & Torsvik (1987)
Mt Hunnenberg Sill, Sweden	6.3	38	166	285	5	Mulder (1971)
Oslo Volcanics, Norway	1.0	47	157	281	5	Van Everdingen (1960)
Ringerike Lavas, Norway	13.0	45	157	281	4	Douglass (1989) in: Van der Voo & Torsvik (2004)
Lunner Dikes, Norway	5.9	53	164	243	5	Torsvik *et al.* (1998)
Sunnhordland dikes, Norway	4.6	50	125	221	5	Walderhaug (1993) in: Van der Voo & Torsvik (2004)
S Poland APWP						
Westphalian/Namurian sediments, Sudetes	7.0	25	178	315	5	Kdziałko-Hofmokl & El-Hemaly (1996)
Upper Caeboniferous volcanics, Sudetes	11.0	39	181	310	4	Westphal *et al.* (1987)
Lower Autunian sediments, Sudetes	3.1	43	173	300	5	Nawrocki (1998)
Cracow Volcanites, component A	4.8	44	175	294	6	this paper
Cracow Volcanites component B	4.1	50	164	269	4	this paper
Zechstein carbonates, Sudetes	5.2	51	164	264	6	Nawrocki (1997, 1998)
Lower Buntsandstein sediments, Holy Cross Mountains	5.0	49	155	250	6	Nawrocki *et al.* (2003)
Muschelkalk carbonates, S Poland	6.0	52	143	243	6	Nawrocki & Szulc (2000)
S Poland – poles presented in Fig. 7						
Visean sediments, Orlej Quarry – pole OC1						
Pole OC1	7.1	46	194	294	4	Nawrocki *et al.* (2005)
– Pole OC2	5.7	45	228	294	4	Nawrocki *et al.* (2005)
– Pole OC3	7.5	44	185	294	4	Nawrocki *et al.* (2005)
Upper Autunian red beds, Płóczki Górne, Sudetes	8.5	47	183	290–270	4	Nawrocki (1998)

Note: The quality factor *Q* was defined according to criterions of Van der Voo (1993); 7 is best.

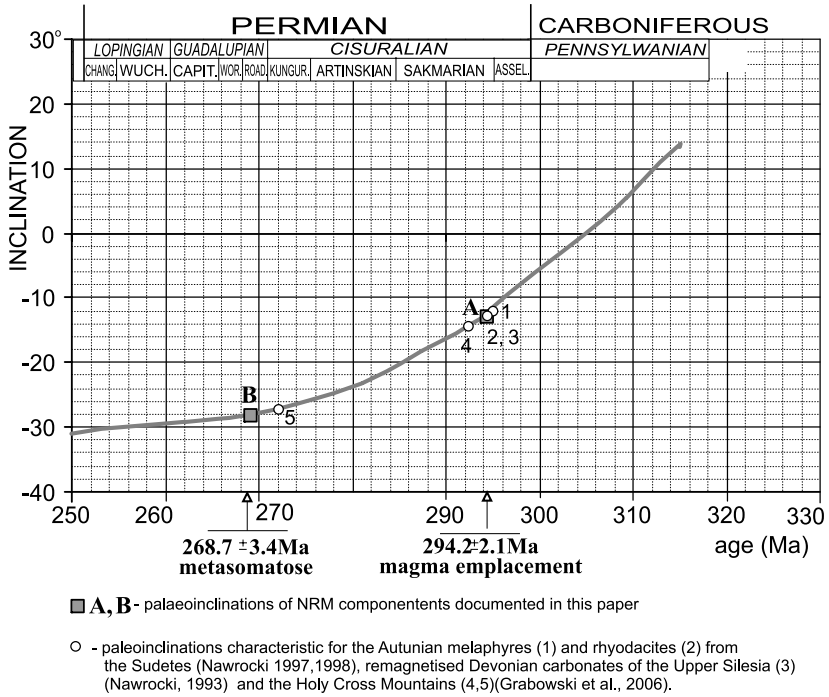


Figure 8. Expected southern Poland Late Carboniferous to Late Permian palaeoinclinations calculated for the geographic coordinates of Cracow (data the same as for APWP of S Poland, Table 4), and inclinations of the NRM components A and B (see text). Palaeoinclinations characteristic for selected Permian rocks from the Sudetes and remagnetized Devonian carbonates of the Upper Silesia and the Holy Cross Mountains are also presented. Time scale was adopted from Gradstein *et al.* (2004).

were inducted most probably by asymmetrical backarc spreading connected with subduction of the Palaeo-Tethys (Stampfli & Borel 2002). Traces of transmission of the Palaeo-Tethys subduction activity can be observed in the North Dobrogea orogen that was affected by last compressional deformations during mid-Permian times (Sandulescu *et al.* 1995). Less probable source of sinistral movements in Central Europe could be related to the collision of Baltica with Kazakhstan as suggested by Mattern (2001).

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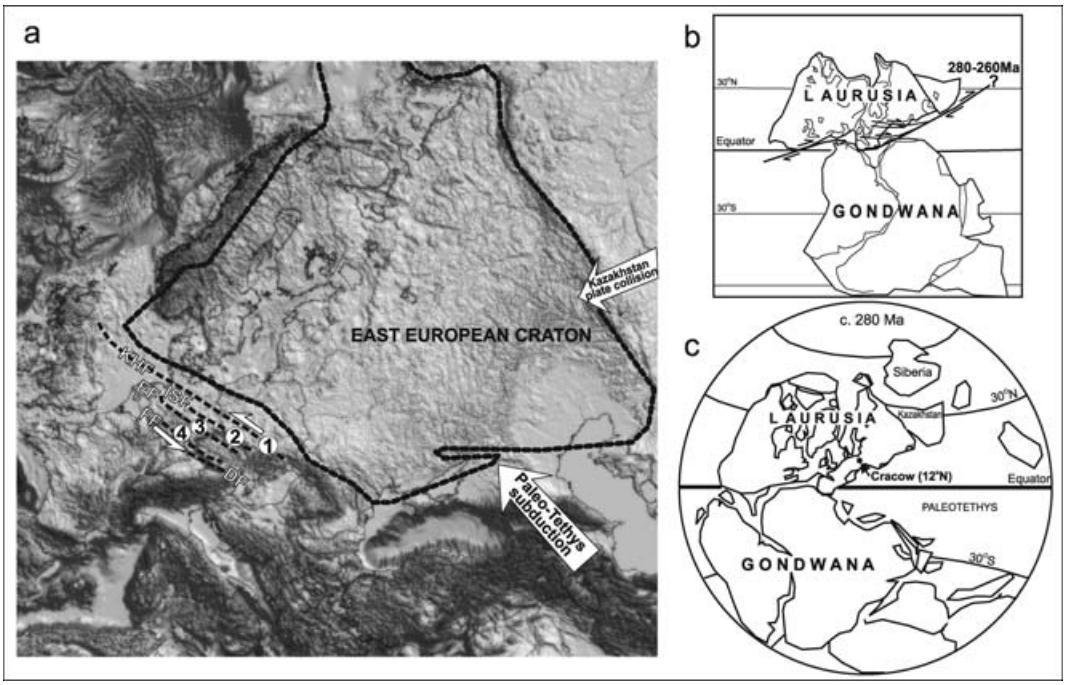


Figure 9. (a) NW-trending faults cutting the Bohemian Massif and southern Poland on the background of shaded relief map of Europe (1 – Cracow-Hamburg Fault Zone; 2 – Intra-Sudetic Fault; 3 – Elbe Fault; 4 – Franconian Fault). Their Permian sinistral tectonic activity and directions of Palaeo-Tethys subduction, and Kazakhstan plate collision are also marked. (b) The mid-Permian transformation of Pangea 'B' to Pangea 'A' along an intra-Pangea dextral megashear system according to the model of Muttoni *et al.* (2003). (c) 280 Ma Pangea reconstruction where Laurussias's position is based poles from S Poland rotated to the Fennoscandian APWP (the running mean pole for 280 Ma: 160°E, 44°N) and Gondwana's position is according to Van der Voo and Torsvik (2004).

The sinistral tectonic activity of major NW-trending faults of Central Europe dos not support the model of Muttonii *et al.* (2003) in which these faults are presented as dextral ones (Fig. 9b; see also Matte 1991). Palaeomagnetic data from the Cracow area and the Sudetes indicate at that time the opposite, that is, sinistral sense of tectonic activity. On the other hand it cannot be definitely excluded that the NW-trending faults of Central Europe were not engaged in dextral megashear of pre-Alpine Europe proposed by Muttoni *et al.* (2003).

A substantial amount of overlap of continental crust in the Early Permian between Gondwana and Laurussia if reconstructed in Pangea 'A' configuration can be significantly reduced after detailed selection of Permian palaeomagnetic poles from Laurussia (Van der Voo & Torsvik 2004). This reduction can be also due to older ages of the Autunian volcanites from post-Variscan Europe than previously assumed. The mean ages of palaeomagnetic poles isolated from the Autunian igneous rocks of western margin of the Bohemian Massif (Soffel & Harzer 1991) and the Sudetes (Nawrocki 1997) is defined in *the Global Palaeomagnetic Database* on 278 and 285 Ma, respectively. Isotope data and the results of comparison of palaeomagnetic inclinations presented here may indicate that at least for the Sudetes and the Cracow area the real age is older than previously assumed. The location of Laurusia at 280 Ma presented in Fig. 9(c) was estimated by interpolation between palaeopole 'A' and 'B'. The overlap of Gondwana's and Laurussia's parts of the Early Permian Pangea 'A' is in this reconstruction unseeable. It should be stressed, however, that Permian palaeomagnetic data for Gondwana are scarce. Gondwana's position is based on the running mean pole of Torsvik and Van der Voo (2002). It does not differ significantly from the location proposed by Rakotosolofo *et al.* (2006) on the basis of the Early Permian palaeomagnetic pole from Peru.

8 CONCLUSIONS

The following conclusions can be drawn from this study.

1. The single-grain U-Pb dating of Cracow igneous rocks indicate the mean age of 294.2 ± 2.1 Ma for the rhyodacite magma emplacement and could indicate younger age (268.7 ± 3.4 Ma) for the metasomatic alterations of these rocks.
2. Two Permian components of magnetizations related to these processes were isolated. The regional APWP characteristic for S Poland constructed on the basis of palaeomagnetic poles defined in the Cracow region and the Sudetes shows a certain departure from the coeval part of the Fennoscandian APWP due to anticlockwise rotations of studied rocks.
3. Detected rotations were caused by the middle Permian sinistral tectonic movements along reactivated prominent Variscan fault of Central Europe.
4. Older than previously assumed ages of the post-Variscan igneous rocks of Central Europe reduce overlap of Gondwana's and Laurussia's parts of the Early Permian Pangea 'A'.
5. Early Permian volcanism and mid-Permian metasomatic activity were coeval with two events of remagnetizations of Devonian carbonates in Poland.

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